

Window Functions

=====

```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY,  
    employee_name VARCHAR(50),  
    department VARCHAR(50),  
    salary DECIMAL(10, 2),  
    hire_date DATE  
);
```

```
INSERT INTO employees (employee_id, employee_name, department, salary, hire_date)  
VALUES  
(1, 'Amit', 'HR', 50000, '2022-01-15'),  
(2, 'Neha', 'HR', 55000, '2023-03-10'),  
(3, 'Suresh', 'HR', 48000, '2021-11-20'),  
(4, 'Rohit', 'HR', 52000, '2022-09-05'),  
(5, 'Raj', 'Finance', 60000, '2021-07-23'),  
(6, 'Ravi', 'Finance', 62000, '2022-09-01'),  
(7, 'Kiran', 'Finance', 58000, '2021-02-14'),  
(8, 'Sunita', 'Finance', 61000, '2023-01-11'),  
(9, 'Priya', 'IT', 70000, '2020-12-02'),  
(10, 'Anjali', 'IT', 67000, '2021-11-19'),  
(11, 'Vikas', 'IT', 69000, '2022-05-20'),  
(12, 'Sanjay', 'IT', 72000, '2023-04-30'),  
(13, 'Meena', 'IT', 68000, '2021-03-15');
```

-- find max salary for each department

```
select department, max(salary) as max_sal  
from employees  
group by department;
```

-- employee_id, employee_name

```
select department, max(salary) as max_sal, employee_id, employee_name  
from employees  
group by department;
```

```
select * from employees e  
join (select department, max(salary) as max_sal  
from employees  
group by department) d  
on e.department = d.department;
```

```
select e.employee_id,  
e.employee_name,  
e.department,  
e.salary,  
e.hire_date  
from employees e  
join (select department, max(salary) as max_sal  
from employees  
group by department) d  
on e.department = d.department and e.salary = d.max_sal;
```

-- I want top 2 employees from each department

-- the below will not work

```
select e.employee_id,  
e.employee_name,  
e.department,  
e.salary,  
e.hire_date  
from employees e  
join (select department, max(salary) as max_sal  
from employees  
group by department) d  
on e.department = d.department  
and e.salary = d.max_sal  
limit 2;
```